PHY202: Problem Set 1

1. Find the real and imaginary parts $u(x, y)$ and $v(x, y)$ of the following functions:

   (i) $\dfrac{2z-i}{iz+2}$, (ii) e^{iz}, (iii) $\sqrt{1+z^2}$.

2. Use the Cauchy-Riemann conditions to find out whether the following functions are analytic or not (use $z = x + iy$):

   (i) $f(z) = y + ix$, (ii) $f(z) = \dfrac{x - iy}{x^2 + y^2}$, (iii) $f(z) = \dfrac{y - ix}{x^2 + y^2}$.

3. A complex function can be written in the polar form $f(z) = R(r, \theta)e^{i\Phi(r,\theta)}$, where $z = re^{i\theta}$. Derive the Cauchy-Riemann conditions in terms of R, Φ and their derivatives w.r.t. r and θ. Use these conditions to determine which of the following functions are analytic:

   (i) $\sqrt{z}$, (ii) $|z|$, (iii) $\ln\ z$.

4. Use the elementary definition of a derivative to check that the following formulae hold for a function of complex variable:

   (i) $\dfrac{d}{dz}\left(\dfrac{f(z)}{g(z)}\right) = \dfrac{gf' - fg'}{g^2}$ $(g(z) \neq 0)$, (ii) $\dfrac{d}{dz}(z^n) = nz^{n-1}$,

   (iii) $\dfrac{d}{dz}(f[g(z)]) = \dfrac{df}{dg}\dfrac{dg}{dz}$.